

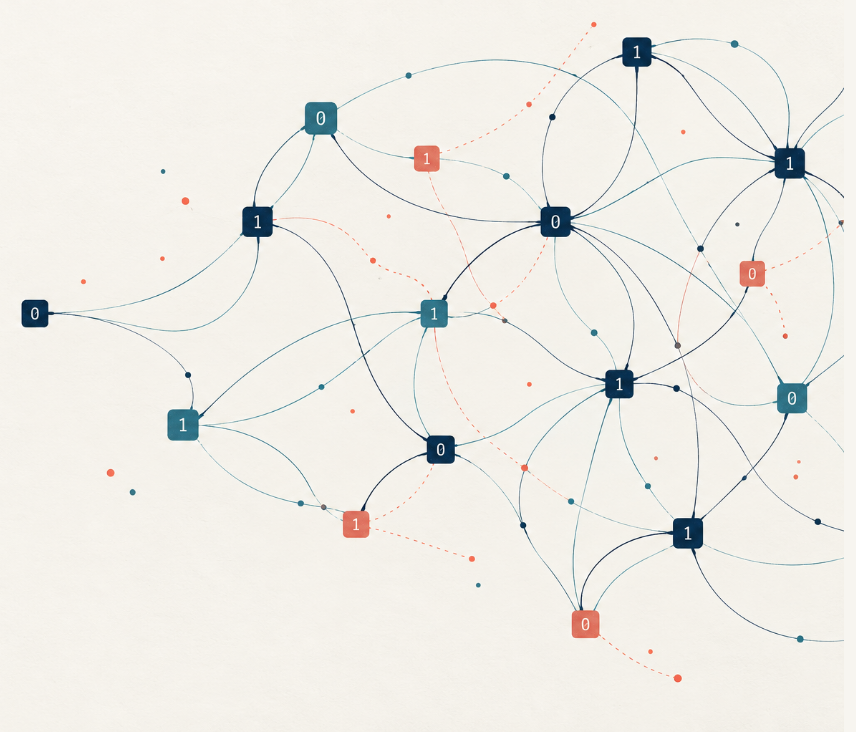
Average-case hardness and the foundations of public-key encryption

Understanding ABW's opening claim

Definitions and worked examples

Research discussion · 6 October 2026

Average-case hardness and ABW



What does “broaden the foundations” mean?

A wider set of assumptions can support PKE.

Phrase	Meaning
Natural combinatorial problems	Local constraints and graphs, studied beyond cryptography
NP-hard optimization	Maximize satisfied equations with three variables each
New average-case assumptions	Difficulty under a specified random sampling rule

Worst-case optimization hardness alone does not establish cryptographic security.

Worst-case and average-case hardness

Worst-case viewpoint

Can one efficient algorithm handle every valid instance?

Difficult inputs may occur rarely under our sampling rule.

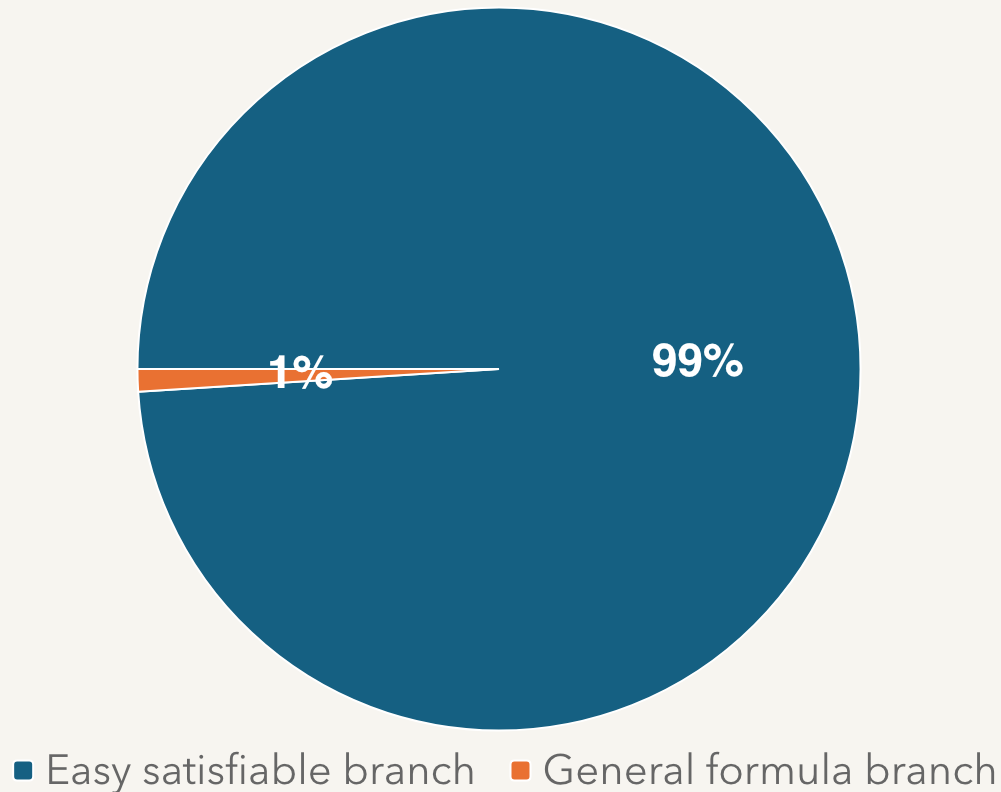
Cryptographic viewpoint

How often can an efficient attacker succeed on generated instances?

The sampling distribution forms part of the assumption.

Worst-case NP-hardness supplies no automatic success bound for the chosen sampler.

Example 1: an easy distribution of SAT inputs



Artificial sampler

99%: obviously satisfiable

1%: general SAT formulas

Always answer “satisfiable”.

Success $\geq 99\%$

SAT remains NP-complete. This particular distribution is easy for the decision task.

Example 2: factoring depends on how inputs arise

Input generation	Small arithmetic example	What the attacker knows
$N = 2p$	$2 \times 1009 = 2018$	2 divides every generated input
$N = pq$, balanced primes	$1009 \times 1013 = 1,022,117$	No factor is deliberately fixed

Both rows ask for factors.

Their generation rules create very different attack opportunities.

Both numerical examples are easy. Large-instance hardness requires an explicit assumption.

Average success comes from a precise experiment

01 Generate an instance

The sampler chooses a public instance I and hidden witness w .

02 Run the attacker

A sees I and may use its own random coins.

03 Measure success

Repeat the experiment according to that same sampling law.

Success means different things for different tasks

Task	Required outcome	Blind baseline
Planted recovery	Output the exact sampled x	2^{-n} for uniform x
Valid-witness search	Output any acceptable witness	Depends on the relation
Balanced distinguishing	Identify which distribution generated I	$1/2$
Optimization	Meet the specified objective or approximation	Depends on the problem

The task and the success threshold belong in every hardness assumption.

Generating an instance can be easy

Generation

Sample x

Compute $y = f(x)$

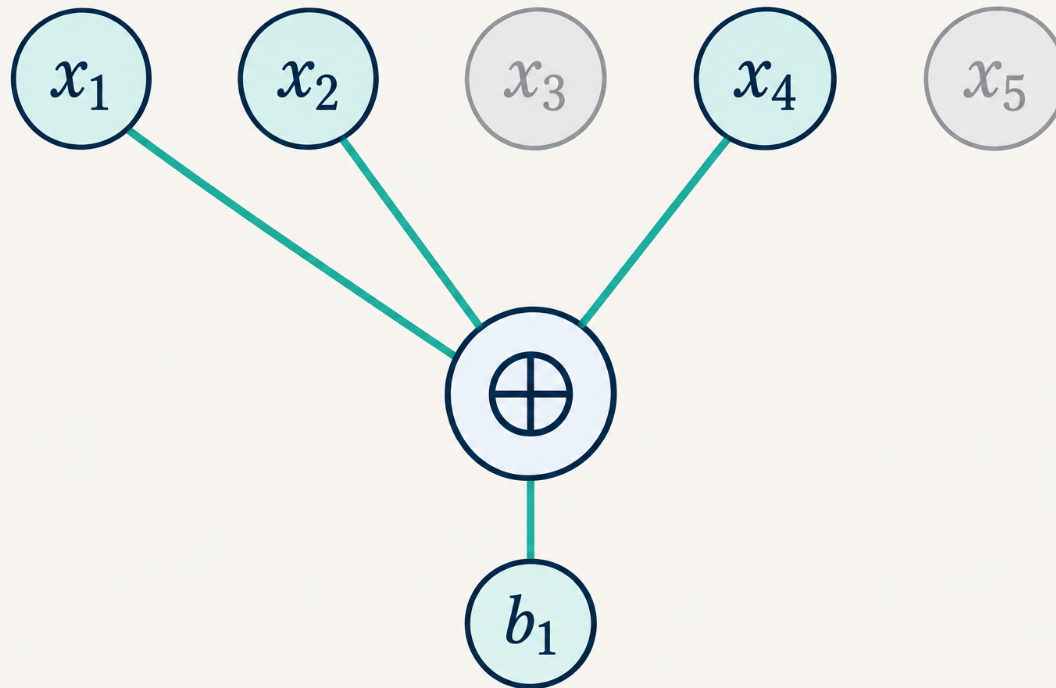
Recovery challenge

Given y

Find a preimage

Easy forward computation can coexist
with conjectured average-case inversion difficulty.

ABW's sparse noisy-equation experiment



$$\mathbf{b} = \mathbf{M}\mathbf{x} + \mathbf{e}$$

$\mathbf{M}$: random rows of weight d
 $\mathbf{x}$: a uniform binary assignment
 $\mathbf{e}_i$: independent output flips

Attacker observes $\mathbf{M}$ and $\mathbf{b}$.

Arithmetic is mod 2 (XOR). The sampling rule specifies n, m, d and ϵ .

Example 3: four noisy parity observations

$$\mathbf{x} = (1, 0, 1, 1)^T$$

Selected variables	Clean parity	Error bit	Observed b_i
$x_1 \oplus x_2 \oplus x_3$	0	0	0
$x_1 \oplus x_2 \oplus x_4$	0	1	1
$x_1 \oplus x_3 \oplus x_4$	1	0	1
$x_2 \oplus x_3 \oplus x_4$	0	0	0

This is one illustrative realization, with one flipped answer.

A perfect fit can miss the planted assignment

Sampled assignment

$$\mathbf{x} = (1, 0, 1, 1)^T$$

One observed equation fails.

An exact fitting assignment

$$\mathbf{z} = (0, 1, 1, 0)^T$$

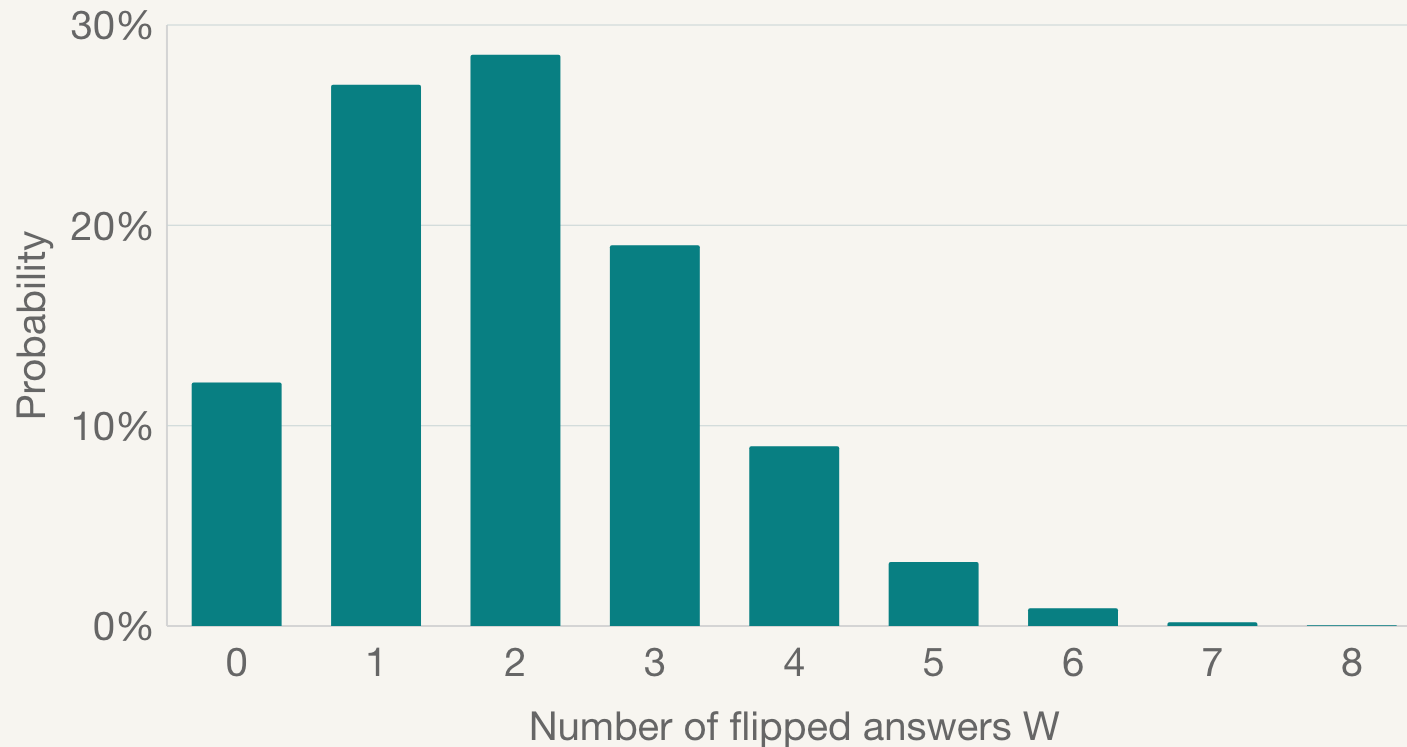
Every observed equation holds.

$$\mathbf{M}\mathbf{z} = \mathbf{b}, \text{ but } \mathbf{z} \neq \mathbf{x}$$

Small algebraic example. A better observed fit need not identify the original hidden assignment.

Expected noise is different from average hardness

Illustration: $m = 20$, $\epsilon = 0.10$



Expected noise

$$E[W] = 2$$

Hardness on average

How often can an attacker recover the hidden assignment?

Calculated distribution, showing 0–8 errors. $E[W] = 2$ does not mean every sample has two errors.

The precise ABW recovery assumption

$$s_A(n) = \Pr[A(M, Mx + e) = x]$$

For every efficient randomized attacker A ,
 $s_A(n) < 2/3$ for all sufficiently large n .

$$d = 3 \quad m = \Theta(n^{1.4}) \quad \varepsilon = \Theta(n^{-0.2})$$

Suitable constants are essential. This statement is an assumption about a specific ensemble.

What the construction theorem establishes

01 Assume a particular average-case problem is hard

The assumption fixes a distribution, task and parameter regime.

02 Construct encryption and analyse attacks

The reduction connects a scheme attack to the underlying problem.

03 Obtain conditional security

The guarantee holds if the assumptions and proof conditions hold.

Assumption, proof and evidence

Type of statement	What it tells us
Hardness assumption	No attacker in the stated class meets the success criterion
Security reduction	A scheme attack would give a relevant problem solver
Restricted lower bound	A particular computational model cannot succeed
Experimental evidence	Tested attacks behaved a certain way at tested sizes

The scope of the evidence determines the scope of the conclusion.

A complete hardness claim for our research

Required field	Question to answer
Distribution	Exactly how do we generate the public observations?
Task	Recover a planted secret, find a witness or distinguish?
Parameters	Which sizes, locality, sample count and noise rate?
Adversary	Which time, memory, sample access and quantum model?
Success bound	What probability or advantage do we rule out?

The problem together with its distribution and success criterion forms the foundation.

Appendix: weak and strong hardness

Strength	Example of a success condition	Interpretation
Weak search hardness	$s_A(n) \leq 1 - \delta(n)$ $\delta(n)$ is noticeable	Every efficient A fails with a noticeable probability
Strong inversion hardness	$s_A(n)$ is negligible	Every efficient A succeeds only rarely
ABW Search3LIN	$s_A(n) < 2/3$ eventually	A constant-threshold premise for its construction

Negligible $v(n)$: eventually $v(n) < 1/p(n)$
for every positive polynomial p .

Appendix: a balanced distinguishing game

P: $(M, Mx + e)$ Q: (M, u)

Choose a hidden fair bit B .
Sample from P or Q according to B .
Ask the attacker to guess B .

Guessing advantage = $|\Pr[\text{correct}] - 1/2|$

Here u is a uniform m -bit vector. Advantage conventions can differ by a factor of two.

Appendix: the worked example in matrix form

M

1	1	1	0
1	1	0	1
1	0	1	1
0	1	1	1

$$Mx = (0, 0, 1, 0)^T$$

$$e = (0, 1, 0, 0)^T$$

$$b = (0, 1, 1, 0)^T$$

$$Mz = b$$

$$z = (0, 1, 1, 0)^T$$

Every row has weight 3. These four equations do not constitute a secure parameter regime.

References and further reading

Applebaum, Barak & Wigderson (2009 / 2010)

Public-Key Cryptography from Different Assumptions
Introduction, Definition 5.1, Assumption 5.2 and Theorems 1–3

Pass & Shelat, A Course in Cryptography

Chapter 2: one-way functions, factoring and hardness amplification
Chapter 3: computational indistinguishability

Arora & Barak, Computational Complexity

Chapter 10: hard-on-average problems and one-way functions

Holenstein & Renner (2005)

Weak public-key encryption and security amplification